

Module 9 Week A — Stretch: Ontology Design

Honors Track. Stretch is for learners who have completed all core assignments, are On Track or Advanced, and are attending consistently. See the Honors Track policy in TalentLMS for full eligibility.

Due: Thursday end of day. Resubmissions are accepted through Saturday of the assignment week.

Prerequisite: Base Lab 9A complete.

What you’ll build

Design a small RDF ontology from scratch — your choice of domain — that exercises the modeling moves you just learned in the lab. The goal is to internalize the patterns by applying them once with no scaffolding.

Pick a domain you find interesting and where you have some intuition about what the entity types and relationships should be. Some examples that work well at this scale: a small board-game catalog, a hiking-trail directory, a personal music-listening log, a small recipe-by-region collection different from the lab.

Requirements

Your ontology must include:

- **~30 triples total.** Enough to be non-trivial; not so many that you’re typing the same pattern 50 times.
- **At least three classes** with at least two instances of each.
- **At least one `rdfs:subClassOf` hierarchy** — for example, `:Mountain rdfs:subClassOf :Terrain`.
- **At least one pair of `skos:prefLabel` + `skos:altLabel`** on a class or instance (a name variant where SKOS disambiguation matters).
- **At least one OPTIONAL-style property** — a property that some but not all instances have. You don’t write `OPTIONAL` in the TTL — you simply leave it off some instances. Document this in the README.

Deliverables

1. `ontology.ttl` — the dataset itself.
2. `README.md` — short writeup covering:
 - The domain and why you chose it.
 - A class diagram or bullet list of the entity types and their relationships.
 - One non-obvious modeling decision and the reason you made it.
 - Where the `skos:altLabel` and `rdfs:subClassOf` appear and what they enable.
3. **Three sample SPARQL queries** in `queries.md` (or as a code block in the README) demonstrating: (i) a subclass-aware query (uses `rdfs:subClassOf*`), (ii) a SKOS-disambiguated query (matches via `skos:prefLabel` OR `skos:altLabel`), (iii) one query of your choice that the domain makes interesting.

Repo setup

This stretch uses a **standalone repo** — not a GitHub Classroom assignment. Create a public repo in your personal GitHub account (suggested name: `aispire-m9a-stretch-ontology`). Push your work. Open a PR on a feature branch if you want code review from a TA; otherwise commit straight to `main`.

Submission

1. Push your repo.
2. **Paste your repo (or PR) URL into TalentLMS → Module 9 → Stretch (Honors) — Ontology design to submit this assignment.**